



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No:- 19	
Name of Student	RAM SHARMA
PRN No.	25070122150
Batch	CSE-B2
Class	PROGRAMMING WITH JAVA
Academic Year & Semester	2nd-year and 3rd-semester
Date of Submission	04/10/2026
Title of Assignment:	Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.
Problem Statement	Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.
Learning objective	To establish database connectivity and retrieve records using JDBC SELECT statements.
Theory	Java Database Connectivity (JDBC) is a standard Java API that enables communication between Java applications and relational databases. It provides a platform-independent mechanism to connect to databases such as MySQL, Oracle, PostgreSQL, and SQL Server. Before executing SQL statements, a JDBC driver is loaded and a connection is established using the DriverManager class. The Connection object represents an active connection to the database, while the Statement object is used to execute SQL queries. The SELECT statement retrieves records from database tables based on

	specified conditions. The retrieved records are stored in a ResultSet object, which allows sequential access to the data. JDBC simplifies database programming and is widely used in desktop, web, and enterprise applications.
Programme	1. Create a Java application to display student records from database using SELECT query => <pre> import java.sql.Connection; import java.sql.DriverManager; import java.sql.ResultSet; import java.sql.Statement; public class E1 { public static void main(String[] args) { try { // Load JDBC driver Class.forName("com.mysql.cj.jdbc.Driver"); // Connect to database Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/jdbc_demo", "root", "admin"); System.out.println("Database connected successfully."); // Create statement Statement st = con.createStatement(); // SELECT query String query = "SELECT * FROM student"; ResultSet rs = st.executeQuery(query); System.out.println("\nStudent Records:"); while (rs.next()) { System.out.println("Roll No: " + rs.getInt("rollno")); System.out.println("Name: " + rs.getString("name")); } } } </pre>

```

        System.out.println("-----");
    }

    rs.close();
    st.close();
    con.close();

    } catch (Exception e) {
        System.out.println("Error: " + e.getMessage());
    }
}
}

```

2. Create a Java application that connects to a database and displays product details using a SELECT query. Show product ID, product name, quantity, and price.

=>

```

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;

public class E2 {

    public static void main(String[] args) {

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/jdbc_demo",
                "root",
                "admin"
            );

            System.out.println("Database connected successfully.");

            Statement st = con.createStatement();

            String query = "SELECT * FROM product";

            ResultSet rs = st.executeQuery(query);

```

```

        System.out.println("\nProduct Details:");

        while (rs.next()) {
            System.out.println("Product ID: " + rs.getInt("id"));
            System.out.println("Product Name: " + rs.getString("name"));
            System.out.println("Quantity: " + rs.getInt("quantity"));
            System.out.println("Price: " + rs.getDouble("price"));
            System.out.println("-----");
        }

        rs.close();
        st.close();
        con.close();

    } catch (Exception e) {
        System.out.println("Error: " + e.getMessage());
    }
}

```

MYSQL QUERY:

```
USE jdbc_demo;
```

```

CREATE TABLE product (
    id INT PRIMARY KEY,
    name VARCHAR(50),
    quantity INT,
    price DOUBLE
);

```

```

INSERT INTO product VALUES
(1, 'Laptop', 10, 55000),
(2, 'Mouse', 25, 800),
(3, 'Keyboard', 15, 1500);

```

```

CREATE TABLE student (
    rollno INT PRIMARY KEY,
    name VARCHAR(50)
);

```

```

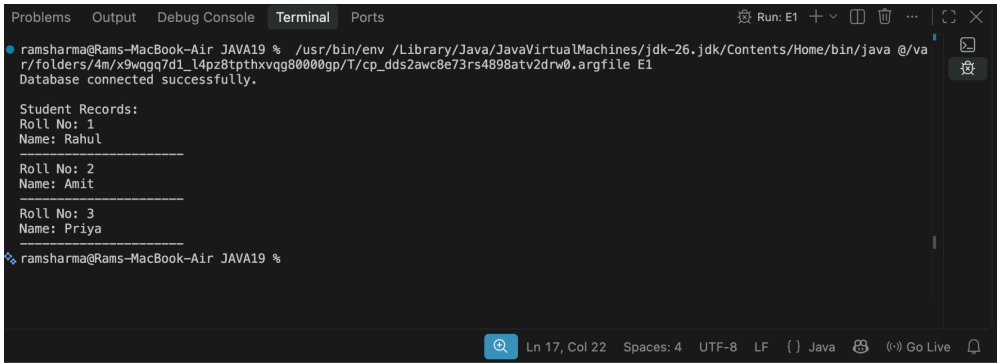
INSERT INTO student VALUES
(1, 'Rahul'),
(2, 'Amit'),
(3, 'Priya');

```

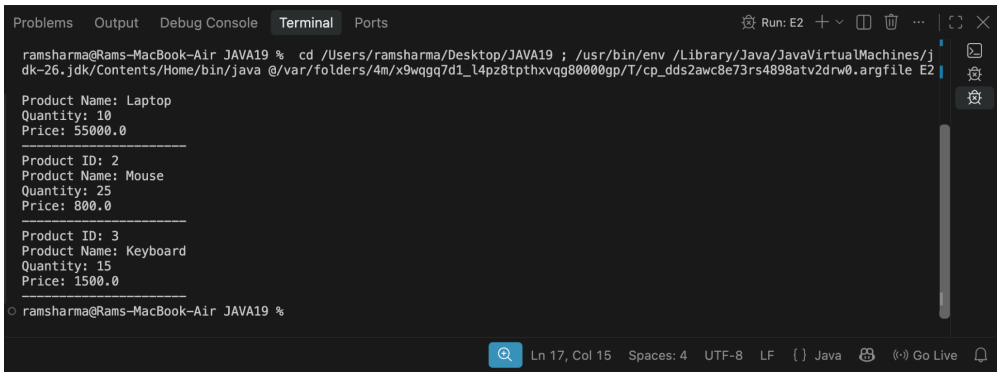
SELECT * FROM student;

Output Screenshots (if applicable)

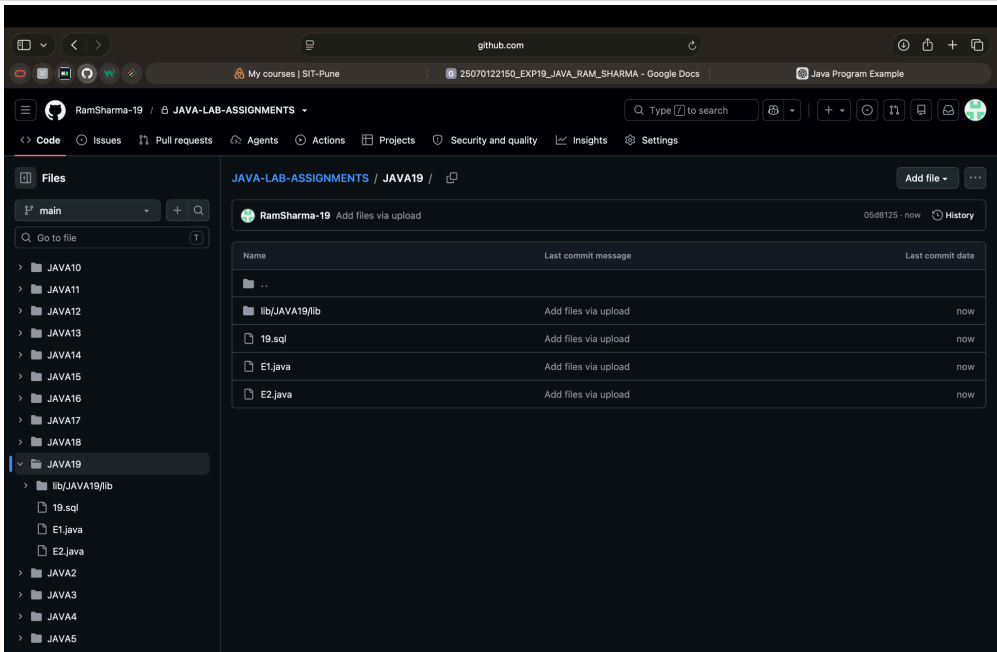
E1 =>



E2 =>



Github Screenshot



Conclusion

In this assignment, JDBC was used to connect a Java program with a MySQL database. The MySQL JDBC driver was loaded and a successful database connection was established using DriverManager. SQL queries were executed from Java to retrieve records from the database. The ResultSet was used to read and display the retrieved data. This experiment helped in understanding the basic process of connecting Java applications with databases and performing **SELECT operations using JDBC**.